

Data Link Layer

Module 07

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sing from Department of computer science Punjabi University patiala today's lecture is on data link layer in this lecture I'll explain you the working of second layer of OSI reference model that is data link layer now let's start in the Osa reference model the layer which is directly on top of physical layer is the data link layer or also called dll the dll is also named as Layer Two or of the OSI model data link layer is one of the most complex layers and has difficult functionalities and responsibilities this layer hides the particulars of underlying hardware and depicts itself to Upper layers as the channel to communicate data link lir works between two communicating hosts which are connected directly in some

sense this direct connection may possibly be broadcast or point to point systems on broadcast network are supposed to be on same link the work of data link layer becomes more difficult when it is dealing with several hosts on single collision domain data link layer converts the incoming data stream into signals bit by bit and to send it over the underlying Hardware at the receiving end data link lay collects the data from underlying Hardware which is in the form of electrical signals build them in an identifiable frame format and pass it over to upper layer data link layer has two sub layers logical link control and media Access Control logical link control also called LLC this sublayer is needed in the case of Point topoint links only it deals with protocols flow control and error

control media access control the Mac sublayer is in charge of deciding which user get to use the broadcast medium at any particular time dll protocols can be implemented either in Hardware or software they can be implemented in either a computer's main CPU or in special purpose Hardware known as network adapter or network interface card n i a common example of the later case is an Ethernet Nic data link lay performs many functions on behalf of upper layer the first function is framing data link layer receives packets from Network layer and encapsulates them into frames the next function is addressing the data link layer provides layer to Hardware addressing system Hardware address is supposed to be unique and is encoded

into Hardware at the time of manufacturing the third function is error control sometimes during transmission signals may encounter problem and the bit get flipped so the detection of errors and recovery of original data is done by this layer the next function is flow control stations present on same link may have different speeds data link layer provides flow control that allows both sender and receiver to exchange data at same speed the last function is multiaccess when host on the shared link tries to transfer the data it has a high probability of collision data link layer provides mechanisms such as CSMA/CD to equip capability of accessing a shared media among multiple systems next we are going to discuss the first function of data link layer that is framing the major

responsibility of data link layer is to make physical link reliable and to do so it breaks the network layer data stream into small blocks this process is called segmentation and in order to form a frame it adds header and frame flag to each and every block and this process is called encapsulation header of the frame generally contains three fields the first field is address field which contains the address of sender and receiver the second field is error detecting code where a checksum of the frame for error detection is stored the third field is control field which stores the additional information to implement protocol functions the receiving data link layer must know the start and the end of a frame according

to the frame flag a good design must make it easy for a receiver to find the start of a new frame while using little of the channel bandwidth we are now going to look at four of the framing methods the first method is byte count this framing method uses a field in the header to describe the number of bytes present in the frame when the receiver's data link layer sees this count it comes to know how many bytes to follow the next method is flag bytes with bit stuffing in this case each frame starts and ends with special bit sequence generally same bit also called a flag bit is used as starting and ending delimiter and this bit sequence is shown in figure as flag sometimes it may happen that the flag bit is present in the data then in

order to solve this problem the sender's data link layer inserts a special bit known as ESC or Escape bit immediately before each accidental flag bit in the data this technique is known as bit stuffing the third method is flag bits 6 minutes, 1 second with bit stuffing as we know that frame flag is a special bit pattern which should not be present anywhere else inside a frame if a special frame flag of 0110 is used at the start and end of the frame and whenever a sender's DLC has five consecutive ones in the data then it automatically adds zero after five consecutive ones and receiver deletes this Z bit that follows five consecutive one bits in the received data this process is called bit stuffing the last method is physical layer coding violations this framing method is used in those networks

in which the encoding on the physical medium contains some redundancy some lenses encode each bit using two physical bits and Manchester coding is normally used in this case the one bit is encoded using 01 and 0 bit is encoded using 10 this type of invalid physical code is used in 802.3 lens standards next is flow control modern networks are aimed to support a wide

variety of hosts and communication mediums suppose a 200 MHz Pentium based host wants to transmit a data to 25 MHz based 80386 host naturally the faster Pentium will drone the slower 80386 with data similarly consider Two Hosts and both of them using an ethernet land but with the two ethernet connected by a 28.8 kbps model link if one host starts transmitting to the other host at ethernet speeds then in this case the model link will quickly

become overburdened in both of these cases flow control is required to speed the data transfer at suitable rate flow control is a technique that informs the sender the amount of data it can transfer before it should wait for an acknowledgement from the receiver the flow of data from sender to receiver should not be allowed to overburden the receiver receiver must also be able to report to the transmitter before its limits are reached and the sender should send less number of frames the limit may well be amount of memory used in order to store the incoming frames or the processing power of receiver the two methods developed for flow control are stop and wait and sliding window first we'll discuss stop and wait stop and wait is the simplest form of flow

control in which a sender transmits a data frame after receiving the data frame the receiver indicates its keenness to accept another frame by sending back an acknowledgement frame acknowledging the frame received earlier the sender must wait until ACK frame is received by it before transmitting the next frame this is also referred to as a request reply mechanism which is very easy to understand and implement but not very efficient in links with fast links this is not a major concern but in links will spend most of their time staying idle especially if a large number of hops are required figure here shows the working of the Stop and wait protocol the blue arrows indicate the sequence of data frames being sent over the link from sender to

the receiver the protocol depends on two-way transmission either half or full duplex to allow the receiver to return acknowledgement frames acknowledging the successful transmission the acknowledgements are shown in green color in this diagram which move back to the original sender a small processing delay is introduced between the reception of the last data frame and the generation of the acknowledgement major drawbacks of this scheme is that only one frame can be transmitted at a time and this leads to inefficiency in the performance if propagation delays longer than the transmission delay next we'll calculate the link utilization in stop and wait protocol let's assume the few things the first is transmission time it is the time taken by a station to

transmit a data frame normalized to a value of one and the second is propagation delay it is a Time taken by a bit to travel from sender to receiver expressed as a if the value of a is less than one the frame is sufficiently long such that the first bit of the frame arrive at the destination before the source has completed transmission of the frame if a is greater than one send a completes transmission of the entire frame before the leading bits of the frame arrive at the receiver the link utilization U is equal to $\frac{1}{1 + 2a}$ where a is equal to propagation time divided by transmission time it is clear from this equation that the link utilization is

strongly dependent on the ratio of the propagation time to the transmission time when the propagation

time is very small as it happens in case of lands the link utilization is good but when the propagation delays are very long in case of uh Satellite Communication the link utilization can become very poor so to improve the link utilization we can use sliding window protocol instead of using stop and weight protocol the next method of flow control is sliding window the simple stop and weight protocol does not work well if multiple data frames are used for single message only one data frame can be in transmission mode at a time in stop and weight protocol if a is greater than one severe inefficiencies result efficiency of the protocol can be improved if multiple frames are transmitted at the same time and also by using the full duplex line the sending station keep track on the

frames by transmitting sequentially numbered frames the sequence number should be of limited size because it occupies a field in the data frame if there are K bits in the header of the data frame then the range of the sequence numbers must be from 0 to $2K$ minus 1 sender keeps a list of sequence numbers that it is authorized to send and this list is called a sender window the maximum size of the sender's window is $2K$ minus 1 the sender is allocated above a space equal to the window size receiver also keeps a window called a receiver window of size $2K$ minus 1 the receiver acknowledges every frame by sending back an acknowledgement frame which includes the sequence number of the expected next frame this also makes a public

announcement that the receiver is ready to receive the next end frames starting with the specified number this scheme is used to acknowledge multiple frames sliding window algorithm provides a flow control for network data transfers TCP protocol uses a sliding window protocol a buffer is placed between the application program and network data flow by the sliding window protocol for TCP the buffer is normally present in the kernel of the operating system the data which is received from the network gets stored in the buffer space from where the application can read data at its own speed as the data is read by the application buffer space gets empty and is ready to accept more data from the network now we are going

to discuss the sender and receiver sliding windows at any point of time the sender is allowed to transmit frames with sequence numbers in a particular range the center window as shown in the figure here the next is receiver sliding window the receiver keeps a window of size one if receiver sees that frame should arrive in a particular order if frames are received out of order then it is discarded and needs to be recent however the receiver window increased by one if a particular frame is received the frame is acknowledged by sending back an AK frame which includes uh the sequence number of the next frame this also openly announces that it is ready to receive the next end frames starting with the specified number this scheme can also be used to acknowledge multiple frames sliding window is beneficial if

local application processes data at the same rate at which it is being transferred if packet size is smaller than the window size then multiple packets can be present in the network because the sender knows that the space exists on the receiver site to store all the packets ideally steady state can be attained where a number of packets in the forward Direction and window announcements in the reverse direction are constantly in traveling when the sender receives new window announcement it transmits more data packets and when data from the buffer is read by the application more window announcements are generated maintaining a sequence of data packets during transfer guarantees the effective use of network resources third function that is error control of

data link layer when the receiver detects an error in the message or packet it then informs the sender to retransmit the particular message or packet the most popular retransmission scheme is known as automatic repeat request or arq the three well-known erq techniques are stop and wait arq go back and erq and selective repeat arq first we will discuss stop and wait arq stop and wait arq is the simplest protocol among all the protocols in which the sender sends a frame and then it waits till a positive acknowledgement or negative acknowledgement is received from the receiver the receiver sends a positive acknowledgement only if the frame is received correctly or else it sends back negative acknowledgement the sender sends a new frame only after

receiving positive acknowledgement otherwise it retransmits the old frame if negative acknowledgement is received this is illustrated in the figure here to deal with the issue of a lost or damaged frame the sender is provided with the timer if acknowledgement is lost the sender transmits the old frame in the figure the second pdu of data is lost during transmission the sender does not know about this loss but starts a timer after transmitting each pdu generally a positive acknowledgement pdu is received before the timer expires in this case no acknowledgement is received and the timer counts down to zero and retransmission of the same pdu is initiated by the sender the sender always starts a timer immediately after transmitting a pdu but the second

transmission receives an acknowledgement pdu before the timer expires finally indicating that the data has now been received by the remote node as the duplicate frame is received by the receiver therefore it is discarded the receiver now can identify that it has received a duplicate frame from the label of the frame and it is discarded to address the issue of lost or damaged frames for example a frame gets corrupted due to noise during the transmission and there is a concept of negative acknowledgement frames the receiver sends back a negative acknowledgement frame to the sender if it finds that the received frame is corrupted when a transmitter receives a negative acknowledgement frame before the timeout of the timer the old frame is transmitted again as shown in the

figure the major advantage of stop and wait protocol is its Simplicity it also requires minimum buffer size however it makes highly inefficient use of communication links particularly when propagation delay is large second arq scheme is go back n arq go back n is one of the most popular arq in which the sender transmits the frames continuously without even waiting for acknowledgement that is why this protocol is called as continuous arq when the frame is received by the receiver it continues to send acknowledgments or negative acknowledgements back to the sender if negative acknowledgement frame is received by the sender it retransmits the frame including all the successive frames as shown in the figure therefore

this protocol is called go back n arq if a frame is lost during transmission then the receiver sends negative acknowledgement frame if there is a long delay in sending the N frame the sender will retransmits the Lost frame after its timer times out if the ACK frame is lost in this case the sender resends the frames after its timer times out as shown in the figure suppose a case of full duplex transmission the receiver sends back piggybacked acknowledgement by using some number in the acknowledgement field of its data frame for example if a 3 bit sequence number is used and suppose that a station sends frame 0 and gets back an ACK and then sends frame 1 2 3 4 5 6 7 and zero and gets another ACK this might either mean

that ACK is a cumulative ACK or all eight frames were damaged this uncertainty can be defeated if the maximum window size is limited to seven that is for a k bit sequence number field it limited to $2^k - 1$ the number n which is equal to $2^k - 1$ specifies how many frames can be sent without receiving acknowledgement if no acknowledgement is received after sending end frames the sender takes the help of a timer after the timer timeout it resumes free transmission the go back n protocol also takes care of damaged frames and damaged acknowledgements this scheme is a little more complex than the previous one but gives much higher throughput the last arq scheme is selective repeat arq the selective repeat arq scheme retransmits only those for which NACKs are received

or for which timer has expired so this is shown in the figure here this is the most efficient method among all the arq schemes but the sender must be more complex so that it can send out of the order frames the receiver also must have storage space to store the post n frames and processing power to reinsert frames in proper sequence so friends this was all about data link layer hope all the concepts are clear to you hope to see you in the next lecture till then goodbye thank you